

Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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5C Upper-Triangular Matrices

Exercise 5C1

Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and T^2 has an upper-triangular matrix with respect to some basis of V , then T has an upper-triangular matrix with respect to some basis of V .

Solution. We claim that the statement is false. Consider the linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ defined by

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

which represents a 90° rotation in the plane. Then T^2 is given by

$$T^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix},$$

which is an upper-triangular matrix with respect to the standard basis of $\mathbb{R}^2$. However, T does not have an upper-triangular matrix with respect to any basis of $\mathbb{R}^2$, since it has no real eigenvalues and thus cannot be represented in upper-triangular form over $\mathbb{R}$. ■

Exercise 5C2

Suppose A and B are upper-triangular matrices of the same size, with $\alpha_1, \dots, \alpha_n$ on the diagonal of A and $\beta_1, \dots, \beta_n$ on the diagonal of B .

- (a) Show that $A + B$ is an upper-triangular matrix with $\alpha_1 + \beta_1, \dots, \alpha_n + \beta_n$ on the diagonal.
- (b) Show that AB is an upper-triangular matrix with $\alpha_1\beta_1, \dots, \alpha_n\beta_n$ on the diagonal.

Axler Note. The results in this exercise are used in the proof of 5.81.

Solution.

- (a) Let $C = A + B$. Since A and B are upper-triangular, for any entry c_{ij} of C where $i > j$, we have

$$c_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0,$$

which shows that C is upper-triangular. The diagonal entries of C are given by

$$c_{ii} = a_{ii} + b_{ii} = \alpha_i + \beta_i,$$

for $i = 1, \dots, n$.

- (b) Let $D = AB$. Since A and B are upper-triangular, for any entry d_{ij} of D where $i > j$, we have the following sum for the (i, j) -th entry of D :

$$d_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Consider a_{ik} when $k < i$. Since A is upper-triangular, $a_{ik} = 0$. Now consider b_{kj} when $k > j$. Since B is upper-triangular, $b_{kj} = 0$. Therefore, for $i > j$, all terms in the sum are zero, and we have $d_{ij} = 0$ for $i > j$, showing that D is upper-triangular.

To see that the diagonal entries of D are $\alpha_1\beta_1, \dots, \alpha_n\beta_n$, we compute the diagonal entries:

$$d_{ii} = \sum_{k=1}^n a_{ik}b_{ki},$$

for the i -th diagonal entry. Since A and B are upper-triangular, then $k > i$ implies $a_{ik} = 0$ and $k < i$ implies $b_{ki} = 0$. Therefore, the only non-zero term in the sum occurs when $k = i$, giving us

$$d_{ii} = a_{ii}b_{ii} = \alpha_i\beta_i,$$

for $i = 1, \dots, n$. ■

Exercise 5C3

Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, \dots, v_n$ is a basis of V with respect to which the matrix of T is upper triangular, with $\lambda_1, \dots, \lambda_n$ on the diagonal. Show that the matrix of T^{-1} is also upper triangular with respect to the basis $v_1, \dots, v_n$, with

$$\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

on the diagonal.

Solution. Let A be the matrix of T with respect to the basis $v_1, \dots, v_n$. Since T is invertible, A is also invertible. Let $A = (a_{ij})$ be the upper-triangular matrix with $\lambda_1, \dots, \lambda_n$ on the diagonal, and let $B = (b_{ij})$ be the matrix of T^{-1} with respect to the same basis. Denote b_j as the j -th column of B . Then we have

$$Ab_j = e_j,$$

where e_j is the j -th column of the identity matrix. In the n th row, we have $\lambda_n b_{nj} = \delta_{nj}$, hence $b_{nj} = 0$ for $j < n$ and $b_{nn} = \lambda_n^{-1}$. Now, consider the $(n-1)$ -th row. We have

$$a_{n-1,n-1}b_{n-1,j} + a_{n-1,n}b_{nj} = \delta_{n-1,j}.$$

For $j < n-1$, we have $b_{n,j} = 0$ and $\delta_{n-1,j} = 0$, so $b_{n-1,j} = 0$. For $j = n-1$, we have $b_{n,n-1} = 0$ and $\delta_{n-1,n-1} = 1$, so $b_{n-1,n-1} = a_{n-1,n-1}^{-1} = \lambda_{n-1}^{-1}$. Assume inductively that for some $k \in \{1, \dots, n-1\}$, we have $b_{ij} = 0$ for all $i > j$ and $b_{ii} = \lambda_i^{-1}$ for all $i \geq k+1$. Now, consider the k -th row. We have

$$a_{kk}b_{kj} + a_{k,k+1}b_{k+1,j} + \dots + a_{kn}b_{nj} = \delta_{kj}.$$

For $j < k$, we have $b_{ij} = 0$ for all $i > j \geq k$, so the left-hand side reduces to $a_{kk}b_{kj} = 0$, hence $b_{kj} = 0$. For $j = k$, we have $b_{ij} = 0$ for all $i > k$, so the left-hand side reduces to $a_{kk}b_{kk} = 1$, hence $b_{kk} = a_{kk}^{-1} = \lambda_k^{-1}$. By induction, we conclude that B is upper-triangular with $\lambda_1^{-1}, \dots, \lambda_n^{-1}$ on the diagonal. ■

Exercise 5C4

Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.

Axler Note. This exercise and the exercise below show that 5.41 fails without the hypothesis that an upper-triangular matrix is under consideration.

Solution. Consider the linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ defined by

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The matrix of T with respect to the standard basis has only 0's on the diagonal, but T is invertible with inverse

$$T^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad \blacksquare$$

Exercise 5C5

Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.

Solution. Consider the linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ defined by

$$T = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Since $T(1, -1) = (0, 0)$, T is not invertible. ■

Exercise 5C6

Suppose $\mathbb{F} = \mathbb{C}$, V is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that if $k \in \{1, \dots, \dim V\}$, then V has a k -dimensional subspace invariant under T .

Solution. The minimal polynomial of T has degree at most $\dim V$, and since $\mathbb{F} = \mathbb{C}$, then the minimal polynomial factors completely into $\dim V$ linear factors. Then T has an upper-triangular matrix with respect to some basis of V by 5.44. Let $v_1, \dots, v_n$ be that basis. By 5.39, $Tv_k \in \text{span}(v_1, \dots, v_k)$ for each $k = 1, \dots, \dim V$. Therefore, $\text{span}(v_1, \dots, v_k)$ is a k -dimensional subspace of V that is invariant under T . ■

Exercise 5C7

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$.

- (a) Prove that there exists a unique monic polynomial p_v of smallest degree such that $p_v(T)v = 0$.
- (b) Prove that the minimal polynomial of T is a polynomial multiple of p_v .

Solution.

- (a) Consider the list $v, Tv, T^2v, \dots, T^{\dim V}v$. Since V is finite-dimensional, this list is linearly dependent. Then there exist scalars $a_0, \dots, a_m$ not all zero such that

$$a_0v + a_1Tv + \dots + a_mT^mv = 0.$$

Here, m denotes the smallest integer such that there exist scalars $a_0, \dots, a_m$ not all zero satisfying the above equation. Let $p_v(x)$ be the monic polynomial after dividing by a_m :

$$p_v(x) = x^m + \frac{a_{m-1}}{a_m}x^{m-1} + \dots + \frac{a_0}{a_m}.$$

Then $p_v(T)v = 0$. Suppose q is another monic polynomial of smallest degree such that $q(T)v = 0$. Then $p_v - q$ is a polynomial of smaller degree than p_v such that $(p_v - q)(T)v = 0$, which contradicts the minimality of p_v .

- (b) Let m_T be the minimal polynomial of T . Then $m_T(T) = 0$, so $m_T(T)v = 0$. By the division algorithm for polynomials, there exist unique polynomials q and r such that

$$m_T = p_v q + r,$$

where $r = 0$ or $\deg r < \deg p_v$. Then

$$0 = m_T(T)v = (p_v q + r)(T)v = p_v(T)q(T)v + r(T)v = r(T)v,$$

since $p_v(T)v = 0$. If $r \neq 0$, then r is a polynomial of smaller degree than p_v such that $r(T)v = 0$, which contradicts the minimality of p_v . Therefore, $r = 0$, and we have $m_T = p_v q$, which shows that m_T is a polynomial multiple of p_v . ■

Exercise 5C8

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and there exists a nonzero vector $v \in V$ such that $T^2v + 2Tv = -2v$.

- Prove that if $\mathbb{F} = \mathbb{R}$, then there does not exist a basis of V with respect to which T has an upper-triangular matrix.
- Prove that if $\mathbb{F} = \mathbb{C}$ and A is an upper-triangular matrix that equals the matrix of T with respect to some basis of V , then $-1 + i$ or $-1 - i$ appears on the diagonal of A .

Exercise 5C9

Suppose B is a square matrix with complex entries. Prove that there exists an invertible square matrix A with complex entries such that $A^{-1}BA$ is an upper-triangular matrix.

Exercise 5C10

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Show that the following are equivalent.

- The matrix of T with respect to $v_1, \dots, v_n$ is lower triangular.
- $\text{span}(v_k, \dots, v_n)$ is invariant under T for each $k = 1, \dots, n$.
- $Tv_k \in \text{span}(v_k, \dots, v_n)$ for each $k = 1, \dots, n$.

Axler Note. A square matrix is called **lower triangular** if all entries above the diagonal are 0.

Exercise 5C11

Suppose $\mathbb{F} = \mathbb{C}$ and V is finite-dimensional. Prove that if $T \in \mathcal{L}(V)$, then there exists a basis of V with respect to which T has a lower-triangular matrix.

Exercise 5C12

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V , and U is a subspace of V that is invariant under T .

- (a) Prove that $T|_U$ has an upper-triangular matrix with respect to some basis of U .
- (b) Prove that the quotient operator T/U has an upper-triangular matrix with respect to some basis of V/U .

Axler Note. The quotient operator T/U was defined in Exercise 38 in Section 5A.

Exercise 5C13

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Suppose there exists a subspace U of V that is invariant under T such that $T|_U$ has an upper-triangular matrix with respect to some basis of U and also T/U has an upper-triangular matrix with respect to some basis of V/U . Prove that T has an upper-triangular matrix with respect to some basis of V .

Exercise 5C14

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an upper-triangular matrix with respect to some basis of V if and only if the dual operator T' has an upper-triangular matrix with respect to some basis of the dual space V' .